

Konstantinos Anagnostopoulos

Mechanical & Computer Engineer

[in linkedin.com/in/konstantinos-anagn](#) [github.com/ntinostempo0](#) [✉ ntinostempo@gmail.com](#)

[📍 Vrioulon 16 Kesariani, Athens, Greece](#) [☎ +30 6984973907](#)



Mechanical and computer engineer with a passion for innovative solutions. **Over 8 years of experience** in automotive engineering, specializing in the design and development of passenger and cargo vehicles. Proficient in **dynamics** and **software development**, with a strong background in structural analysis, CAD, control systems, optimization, mobile, desktop and web app development. Adept at leading cross-functional teams and managing complex projects from concept to completion. Committed to continuous learning and staying abreast of industry advancements to drive excellence in engineering design and performance.

Knowledge

- Programming:** Applications (Desktop, Mobile, Server Web, Client Web), Embedded Systems, Machine Learning, Data Analysis, Model Based Systems Engineering, Mechanical/Mathematical Solvers
- Dynamics:** Ground Vehicles, UAVs, Multibody Dynamics, Thermal Dynamics, Control Systems, Robotics
- Structural:** FEA (Linear, Nonlinear, Explicit), Fatigue, Modal
- Math:** ODEs, PDEs, Numerical Methods, Optimization, Frequency Domain Analysis

Job Experience

Mechanical & Software Engineer - Petros Petropoulos AEBE - Ecoshift (Athens)

Jan 2022 - now

Web/Desktop/Mobile Apps

Mechanical Design

Dynamics

FEA

Automotive CRM & After-Sales Management Platform

Designed and deployed a centralized web portal to streamline vehicle lifecycles, partner communications, and post-sale support workflows.

Network Management

Lifecycle & Ticketing

DevOps & Deployment

Python

Django

PostgreSQL

REST API

DigitalOcean / CI-CD

Full Stack

Smart Electric Scooter Mobile App (iOS & Android)

Developed a cross-platform mobile application enabling keyless vehicle control and peer-to-peer sharing capabilities.

Wireless Unlocking

Digital Key Sharing

C#

.NET MAUI

BLE

Firebase

Google Cloud

Internal Assembly Line ECU Setup Tool

Engineered a desktop application used directly in the production facility for hardware configuration.

Hardware Provisioning

Serial Communication

C#

WPF

Serial Communication

Inno Setup / Installer

Electric Scooter Mechanical Design

CAD Design, structural engineering, kinematics optimization, and electrical integration for core vehicle chassis components.

Structural & Dynamics

Electrical Systems

SolidWorks

Autodesk Inventor

2D Drawings / GD&T

Kinematics

FEA

• Advanced Structural Dynamics & Multiphysics Optimization

Executed numerical analysis, fatigue life evaluation, and structural optimization projects for complex mechanical systems.

- **Vibration & Fatigue:** Conducted frequency domain analysis and random response simulations to assess structural durability under dynamic loads. Topology optimization under modal analysis for increased frequency of the first mode.
- **Optimization Frameworks:** Applied genetic algorithms and gradient-based methods for topology, shape, MBD and other multi-physics optimizations.

Implicit / Explicit FEA

Topology / Shape Optimization

Multibody Dynamics (MBD)

Optimization Algorithms

Modal

Dynamics Engineer - [GIVE](#) (Athens)

Sep 2018 - Dec 2021

FEA Development

Simulation

Modeling

Mechanical Design

Dynamics

FEA

• Digital Twin & Systems Engineering for Tilted-Rotor UAV - [📺 Demo](#)

Developed a comprehensive Digital Twin framework using Model-Based Systems Engineering (MBSE) and multidisciplinary optimization methods to simulate and streamline tilted-rotor UAV design.

- **Component Selection Tool:** Integrated optimization algorithms to evaluate and select optimal propulsion and aerodynamic components.
- **Interactive Dashboard:** Built dynamic web interfaces to visualize real-time simulation data, flight metrics, and performance trade-offs.
- **Project Management:** Coordinated project lifecycles, requirement tracking, and milestone delivery following MBSE standards.

Python

Plotly Dashboard

xFoil

Nonlinear ODEs

Lagrangian Mechanics

MBSE

• Electrical model of battery cell with Kalman filter for SOC estimation

Curve fit, Kalman filter implementation and SOC estimation of a Li-ion battery cell using Python and MATLAB. Scaled up to battery pack level.

- **Battery modeling:** Developed an equivalent circuit model (2nd order) of a Li-ion battery cell to capture its dynamic behavior.
- **Curve fitting:** Performed curve fitting to model the battery's voltage characteristics under various operating conditions based on experimental data.
- **Kalman filter implementation:** Implemented a Kalman filter for state-of-charge (SOC) estimation, improving the accuracy of the battery model.

Genetic Algorithms

Nonlinear 2nd order ODEs

Kalman Filter

MATLAB/Simulink

• Leaf Spring Designer desktop app -

A PyQt desktop application for the design and analysis of heavy vehicle leaf springs. The app provides a user-friendly interface for inputting design parameters, performing calculations, and visualizing results. Results can be corrected by using a neural network trained on real test data.

- **Finite Elements:** Implemented 2D beam, spring, gap and rod elements for accurate structural analysis.
- **Neural Network Integration:** A neural network implemented from scratch to correct simulation results based on real-world test data.

PyQt

Non-Linear FEA

2D Finite Elements

Newton-Raphson (quasi-static)

Classification

Neural Networks

• 1D Thermal Analysis software development - [🔗 Repository](#)

A Python module for the thermal analysis of mechanical systems. The script implements a 1D finite element method to solve the heat conduction and convection transient equations and provides temperature distribution results for various boundary conditions.

- **Implementation:** The module is implemented in Python and uses the NumPy and SciPy libraries for numerical computations and Plotly for data visualization. The solver can be selected between implicit, explicit and Crank-Nicolson methods which developed for this project.
- **Testing:** Tested with commercial 3D finite element software.

Python

Heat Equation

Implicit/Explicit

Crank-Nicolson

• Electric Scooter Mechanical Design

CAD Design of various mechanical components, structural engineering and suspension optimization for rider's comfort and traction. Multibody dynamics analysis of the vehicle in several road conditions and inputs.

- **Dynamics:** Designed and simulated suspension geometry, wheels, side-stand, and center-stand for optimal stability and durability.
- **Structural:** Authored complete electrical schematics and designed the full vehicle wiring loom (harness).

Autodesk Inventor

HyperMesh/Optistruct

MotionView/MotionSolve

Multi-Obj Optimization

Frequency Response

• Several projects on vehicle dynamics and structural analysis of mechanical systems.

Conducted in-depth dynamics and kinematics analysis, static calculations, and structural analysis of various mechanical systems.

- **Dynamics:** Simulation of durability tests. Suspension optimization for heavy load vehicles. Flexible body analysis in hydraulic cargo boxes for fatigue analysis. Dynamics of innovative wheelchair with active control angle adjustment in passenger's seat.
- **Structural:** Simulation of small wind turbine components for structural integrity and conduction of report for certification authority. Topology optimization in frequency response analysis for engine subframes.
- **Excel:** Development several calculation templates for mechanical design and data analysis in Excel spreadsheets with VBA macros and buttons.
- **Hand Calculations:** Development of manual calculation methods for verifying simulation results and designing mechanical components. Development of several templates for machine elements calculations (bolts, gears, bearings, springs, etc.) mostly in Excel.

HyperMesh/Optistruct

MotionView/MotionSolve

Control Systems

Multi-Obj Optimization

Excel/VBA

Personal Projects

Mobile Apps

Web Development

• Automatic save of car location for Android - [▶ Play Store](#)

Developed in Flutter only for Android, this app automatically saves the car's location.

- **How it works:** The app works in background (background service and background location permission is required) and automatically saves the car's location when the Bluetooth connection is lost.

Flutter

Android Application

Bluetooth

Location Services

Background Service

• Web Application for splitting expenses - [🔗 Repository](#) - [🌐 Live Demo](#)

Python Django web application for splitting expenses between friends. The app allows users to create groups, add expenses, and calculate the amount each person owes or is owed. An authenticated user account can be provided upon request.

- **Host Server:** It is hosted in local home pc (ubuntu server) using Tailscale funnel service.
- **Database:** SQLite database is used to store user data and expenses.
- **Authentication:** User authentication is implemented using Django's built-in authentication system.
- **Save as App:** The web application can be saved as a mobile app in both Android and iOS devices as Progressive Web Apps (PWAs), allowing users to install it on their devices by "Add to Home Screen" option.

Python

Django

Tailscale Funnel

Ubuntu Server

SQLite

Progressive Web App (PWA)

Skills

Languages

Greek ★★★★★
English ★★★★★☆
German ★★☆☆☆☆

Programming Languages

Python ★★★★★
C/C++ ★★★★★
C# ★★★★★
HTML/CSS/JS ★★★★★☆
Fortran ★★★★★☆
Java ★★★★★☆

Frameworks & More

Django ★★★★★
MAUI ★★★★★
Flutter ★★★★★
SQL & NoSQL ★★★★★☆
Google Cloud ★★★★★☆
VBA ★★★★★☆
SKLearn ★★★★★☆
TensorFlow/Keras/Pytorch ★★★★★☆

Programming Software

Visual Studio ★★★★★
VS Code ★★★★★
Jupyter Notebook ★★★★★
Git(Hub) ★★★★★
Linux ★★★★★☆
Docker ★★★★★☆
DBeaver ★★★★★☆

Mechanical Software

SolidWorks & PDM ★★★★★
Inventor ★★★★★
Altair HyperWorks ★★★★★
Matlab/Simulink ★★★★★☆
Webots ★★★★★☆
ROS ★★★★★☆
Gazebo ★★★★★☆

Higher Education

Undergraduate, Informatics & Telecommunication, National and Kapodistrian University of Athens Jan 2025 - now Specialization: Computer Engineering
Integrated Bachelor & Master, Mechanical Engineering, National Technical University of Athens. Sep 2012 - Sep 2018 Specialization: Ground & Aerial Vehicles Grade: 7.4/10. Thesis: <i>Car Accident Reconstruction using energy methods.</i> Published Paper: Anagnostopoulos, K., et al. "Car accident reconstruction using energy methods." <i>International Journal of Crashworthiness</i> 25.1 (2020): 1-12.

Certifications

HyperMesh 2017 Intermediate Examination  April 2019 Issued by Altair University.	Summer School on Industrial Product Design and Concept Development  July 2019 Issued by University of Aegean.	IBM Data Science  August 2020 Issued by IBM/Coursera.
Machine Learning & Self-Driving Cars: Bootcamp with Python  November 2022 Issued by Udemy.	Introduction to Artificial Intelligence and to Prediction & Estimation Algorithms  June 2023 Issued by NKUA.	Getting Started with Accelerated Computing in Modern CUDA C++  February 2026 Issued by NVIDIA.

References

Dr. Elias Nassiopoulos  linkedin.com/in/elias-nassiopoulos  i.nassiopoulos@gmail.com	Dr. Dimitrios Koulocheris  personal page  dbkoulva@mail.ntua.gr
--	---

Other Information

Memberships  Member of the Technical Chamber of Greece (TEE – TCG)
Personal Page  https://ntinostempo0.github.io/Resume/